

WASP-52b

R-1899

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_150827 · created 2026-08-03T14:04:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457261.7505 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.063
Transit depth (Rp/Rs)^2	1.64 +/- 1.6 [%]
Orbital Inclination (inc)	81.82 +/- 2.77
Airmass coefficient 1 (a1)	21.225 +/- 0.021
Airmass coefficient 2 (a2)	-2.58 +/- 1.08
Scatter in the residuals of the lightcurve fit is	0.3 %
Best Comparison Star	#2 - [508, 60]
Optimal Aperture	52.15
Optimal Annulus	132.1
Transit Duration (day)	0.016 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.128 ± 0.063 vs 0.1646 (deviation 0.58σ)
Duration fit vs published	0.016 d vs 0.0758 d (deviation 3.99σ)
Mid-time O–C	3.3 min
Depth SNR	2.0
Residual scatter	0.3 %

Light curve

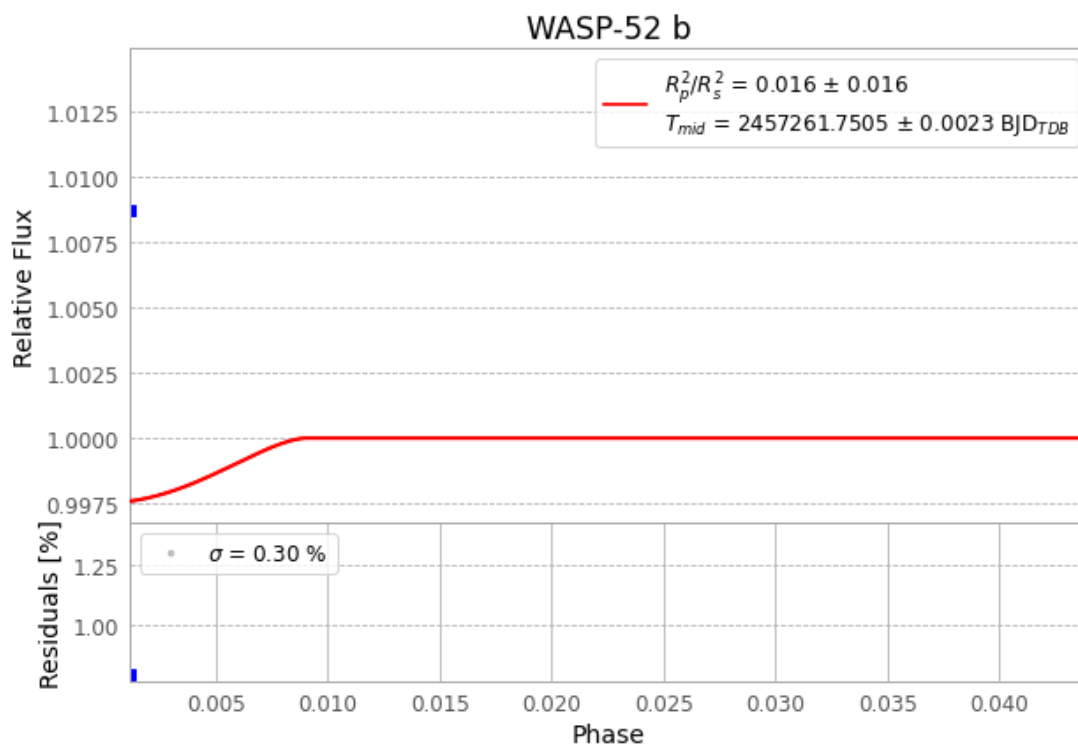


Figure 1 — FinalLightCurve_WASP-52 b_2015-08-26.png (SHA-256 9de57e5ab0979f60...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [551, 309], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 83b80594f234e8f2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52150827040312.FITS → WASP-52150827075112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-150827.log (fc819f6a9e7e...), FinalLightCurve_WASP-52 b_2015-08-26.png (9de57e5ab097...), FinalLightCurve_WASP-52 b_2015-08-26.csv (070c694e1bd3...)

Compute resources

Wall time 996.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 14:04Z.